MICROSOFT AZURE | AZ-104 ADMINISTRATOR ASSOCIATE

Official Exam Preparation Guide · Interview Questions & Answers

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Az

104

300 Interview Questions & Answers

Comprehensive Exam Prep · Azure Administrator Associate

300	13	Expert	A4
Questions	Sections	Level	Format

Identity & AD · Virtual Machines · Networking · Storage · Security · Monitoring · Databases

WHAT YOU WILL LEARN:

- ✔ Identity, Access & Azure AD — Users, Roles, MFA, Conditional Access
- ✔ Virtual Machines, Scale Sets, Availability — IaaS & Compute
- ✔ VNet, NSG, Load Balancer, VPN & ExpressRoute — Networking
- ✔ Blob, File, Queue Storage & Replication — Azure Storage
- ✔ Security, Key Vault, Defender for Cloud — Protection & Compliance

Prepared by Akhilesh Chaudhari

Senior IT Engineer | @akhilesh.webdev | akhileshchaudhari.com

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■ **How to use:** Each section has a **blue question box** and a **green answer box**. Read the question, think of your answer, then verify. Covers all official AZ-104 exam domains. Written in simple language for fast learning.

1 Azure Identity & Access Management

Q1. What is Azure Active Directory (Azure AD)?

✔ **Answer:** Azure Active Directory (Azure AD) is Microsoft's cloud-based identity and access management service. It helps employees sign in and access resources like Microsoft 365, Azure portal, and thousands of SaaS applications. It acts as the 'gatekeeper' for your cloud environment.

Q2. What is the difference between Azure AD and traditional Windows Active Directory?

✔ **Answer:** Traditional Windows AD is used for on-premises environments (domain controllers, group policy), while Azure AD is cloud-based and uses modern protocols like OAuth 2.0, SAML, and OpenID Connect. Azure AD has no concept of OUs, forests, or Kerberos — it's designed for web-based authentication.

Q3. What is Azure AD Connect and why is it used?

✔ **Answer:** Azure AD Connect is a tool that synchronizes on-premises Active Directory identities to Azure AD. This enables hybrid identity — users can use the same username/password for both on-premises and cloud resources (Single Sign-On). It supports password hash sync, pass-through authentication, and federation.

Q4. What are Azure AD Roles?

✔ **Answer:** Azure AD roles define what administrative actions a user can perform in Azure AD. Examples include Global Administrator (full control), User Administrator (manage users), Billing Administrator, and Security Administrator. These are different from Azure RBAC roles which control Azure resource access.

Q5. What is Role-Based Access Control (RBAC) in Azure?

✔ **Answer:** RBAC is a security model that controls who can do what on which Azure resources. You assign a Role (like Reader, Contributor, Owner) to a Security Principal (user, group, service principal) at a specific Scope (subscription, resource group, resource). It follows the principle of least privilege.

Q6. What are the built-in Azure RBAC roles?

✔ **Answer:** Key built-in roles are: Owner (full access + can delegate), Contributor (manage resources but can't grant access), Reader (view only), and User Access Administrator (manage access only). There are 70+ built-in roles for specific services like Virtual Machine Contributor, Storage Blob Data Contributor, etc.

Q7. What is a Service Principal in Azure?

✔ **Answer:** A Service Principal is a security identity used by applications, services, or automation tools to access specific Azure resources. Instead of using a personal account, apps use a service principal with specific permissions — similar to a 'service account' in traditional IT. It's created when you register an app in Azure AD.

Q8. What is a Managed Identity in Azure?

✔ **Answer:** A Managed Identity is a special type of service principal that is automatically managed by Azure. You don't need to manage credentials — Azure handles rotation automatically. There are two types: System-Assigned (tied to a specific resource, deleted when resource is deleted) and User-Assigned (standalone, can be shared across resources).

Q9. What is Multi-Factor Authentication (MFA) in Azure?

✔ **Answer:** MFA requires users to verify their identity using two or more methods: something you know (password), something you have (phone/authenticator app), or something you are (biometric). In Azure, MFA can be enforced via Conditional Access policies or per-user MFA settings. It greatly reduces account compromise risk.

Q10. What is Conditional Access in Azure AD?

✔ **Answer:** Conditional Access is an Azure AD Premium feature that applies access policies based on conditions. For example: 'If a user is outside the corporate network AND tries to access sensitive app, require MFA.' You can block access, require compliant devices, or enforce MFA based on user, location, device, app, or risk signals.

Q11. What is Azure AD B2B?

✔ **Answer:** Azure AD B2B (Business-to-Business) allows you to invite external users (partners, vendors) to collaborate using their own identity. Guest users get access to your apps without needing a separate account. They sign in with their existing work, school, or personal Microsoft account. Access can be limited to specific applications.

Q12. What is Azure AD B2C?

✔ **Answer:** Azure AD B2C (Business-to-Consumer) is a customer identity management solution. It lets your external customers sign in to your apps using social identities (Google, Facebook) or custom accounts. Unlike B2B, B2C is for millions of end-user customers, not business partners. It's a separate Azure AD tenant.

Q13. What is Privileged Identity Management (PIM)?

✔ **Answer:** PIM provides just-in-time privileged access to Azure AD and Azure resources. Instead of having permanent admin roles, users can activate elevated privileges only when needed, for a limited time, with approval. This reduces the risk of standing privileged accounts. PIM requires Azure AD Premium P2.

Q14. What is Identity Protection in Azure AD?

✔ **Answer:** Azure AD Identity Protection uses machine learning to detect risky sign-ins and compromised accounts. It detects risks like leaked credentials, impossible travel, anonymous IP usage, and atypical locations. You can configure risk-based Conditional Access policies that automatically respond to detected threats.

Q15. What is an Azure AD Tenant?

✔ **Answer:** An Azure AD Tenant is a dedicated instance of Azure AD that an organization gets when it signs up for Microsoft cloud services. It represents your organization and is identified by a unique domain (e.g., yourcompany.onmicrosoft.com). All users, groups, apps, and policies exist within a tenant.

Q16. What is the difference between Authentication and Authorization?

✔ **Answer:** Authentication (AuthN) is verifying WHO you are — proving your identity (e.g., logging in with username/password). Authorization (AuthZ) is determining WHAT you can do — what resources or actions you're permitted after being authenticated. Azure uses Azure AD for authentication and RBAC/Conditional Access for authorization.

Q17. What are Azure AD Groups?

✔ **Answer:** Azure AD Groups are used to manage access for multiple users at once. There are two types: Security Groups (used for assigning permissions to resources) and Microsoft 365 Groups (for collaboration with email, Teams, SharePoint). Groups can be assigned-membership or dynamic (auto-based on user attributes).

Q18. What is Dynamic Group Membership?

✔ **Answer:** Dynamic group membership automatically adds/removes users or devices based on attribute rules you define. For example: 'All users with Department = Finance' — as soon as a user's department changes, group membership updates automatically. This requires Azure AD Premium P1 license.

Q19. What is an App Registration in Azure AD?

✔ **Answer:** App Registration allows developers to register their application with Azure AD to enable identity features. When you register an app, it gets a unique Application (Client) ID and can be configured with permissions, redirect URIs, and certificates/secrets. This is required for apps that use Azure AD for authentication (OAuth/OIDC).

Q20. What is SSPR (Self-Service Password Reset)?

✔ **Answer:** SSPR allows users to reset their own passwords without contacting IT helpdesk. Users can verify identity through methods like email, phone, authenticator app, or security questions. It reduces helpdesk tickets and improves user productivity. SSPR can be enabled for all users or specific groups in Azure AD.

Q21. What is an Access Review in Azure AD?

✔ **Answer:** Access Reviews are periodic reviews to ensure users still need access to resources. Managers or resource owners review group memberships or app assignments and approve/deny access. This is a governance feature to maintain least privilege over time. Access Reviews require Azure AD Premium P2.

Q22. What is a Guest User in Azure AD?

✔ **Answer:** Guest users are external identities invited to your Azure AD tenant for collaboration (B2B). Their user type shows as 'Guest' unlike internal 'Member' users. Guests have limited default permissions and can be restricted in what they can see/do in your tenant. Access is managed via invitations.

Q23. What is Azure AD Domain Services (Azure AD DS)?

✔ **Answer:** Azure AD DS provides managed domain services like domain join, LDAP, Kerberos, and NTLM — without needing to deploy or manage domain controllers. It's useful when you have legacy apps that need traditional AD features but you want to move them to Azure without maintaining on-prem DCs.

Q24. What is the difference between Azure AD Free and Premium tiers?

✔ **Answer:** Azure AD Free includes basic features like user/group management and SSO for 10 apps. Premium P1 adds Conditional Access, dynamic groups, hybrid identity, and self-service. Premium P2 adds Identity Protection, PIM, Access Reviews, and entitlement management. Higher tiers provide stronger security controls.

Q25. What is Entitlement Management in Azure AD?

✔ **Answer:** Entitlement Management automates identity governance workflows. You create 'Access Packages' — bundles of resources (groups, apps, SharePoint sites) — and define who can request them, approval policies, and expiration. It simplifies managing access for employees, partners, and guests at scale. Requires Azure AD Premium P2.

2 Azure Virtual Machines

Q26. What is an Azure Virtual Machine (VM)?

✔ **Answer:** An Azure VM is an on-demand, scalable computing resource that gives you the flexibility of virtualization without needing to buy and maintain physical hardware. You choose the OS (Windows or Linux), VM size (CPU/RAM), storage, and networking. VMs are IaaS (Infrastructure as a Service) resources.

Q27. What are VM Size families in Azure?

✔ **Answer:** Azure offers several VM families: General Purpose (B, D, Dsv series) for balanced workloads; Compute Optimized (F series) for high CPU; Memory Optimized (E, M series) for in-memory databases; Storage Optimized (L series) for high disk I/O; GPU VMs (N series) for graphics/AI; and High Performance Compute (H series).

Q28. What is an Availability Set?

✔ **Answer:** An Availability Set groups VMs to protect against hardware failures and planned maintenance. VMs in an availability set are spread across Fault Domains (separate physical racks/power/network) and Update Domains (VMs rebooted at different times during maintenance). It provides up to 99.95% SLA.

Q29. What is an Availability Zone?

✔ **Answer:** Availability Zones are physically separate datacenters within an Azure region, each with independent power, cooling, and networking. Deploying VMs across zones protects against entire datacenter failures. Azure guarantees 99.99% SLA for VMs deployed across two or more zones. Zones are not available in all regions.

Q30. What is the difference between Availability Sets and Availability Zones?

✔ **Answer:** Availability Sets protect within a single datacenter (against rack/hardware failures) and provide 99.95% SLA. Availability Zones span separate datacenters within a region and provide 99.99% SLA with higher protection. Zones protect against larger-scale failures. For critical workloads, zones provide stronger guarantees.

Q31. What is Azure VM Scale Sets (VMSS)?

✔ **Answer:** VM Scale Sets let you create and manage a group of identical, load-balanced VMs that can automatically scale in/out based on demand or schedule. They're used for applications needing high availability and scalability. Scale sets can integrate with Azure Load Balancer or Application Gateway.

Q32. What is the difference between stopping (deallocating) and stopping a VM without deallocating?

✔ **Answer:** Stopping with 'Deallocate' releases the underlying hardware, so you stop being charged for compute. The VM's state is saved on disk, and the public IP may change. Stopping without deallocating keeps the VM on dedicated hardware (you're still charged for compute) and preserves the IP. Deallocated is the recommended way to save costs.

Q33. What are VM Disks in Azure?

✔ **Answer:** Azure VMs use managed disks. Each VM has an OS Disk (required, contains OS, usually 128GB+), optionally a Temporary Disk (local SSD, for page files, data is lost on resize/restart), and Data Disks (attached for application data). Disk types: Ultra Disk, Premium SSD, Standard SSD, Standard HDD.

Q34. What is Azure Managed Disk?

✔ **Answer:** Managed Disks are Azure-managed VHD files stored in Azure Storage. You don't need to manage storage accounts — Azure handles durability, replication, and availability. They support snapshots, encryption at rest, and access control. Most VMs should use managed disks over unmanaged (storage account-based) disks.

Q35. What is VM Disk Encryption in Azure?

✔ **Answer:** Azure Disk Encryption uses BitLocker (Windows) or dm-crypt (Linux) to encrypt OS and data disks. Encryption keys are stored in Azure Key Vault. Server-Side Encryption (SSE) is enabled by default for all managed disks using platform-managed keys. Customer-managed keys (CMK) give you control over encryption keys.

Q36. How do you connect to a Windows VM in Azure?

✔ **Answer:** Connect to a Windows VM using RDP (Remote Desktop Protocol) on port 3389. You can use the Azure portal's 'Connect' button, which downloads an RDP file. For security, it's recommended to use Azure Bastion (browser-based RDP without exposing port 3389) or configure Just-In-Time VM Access via Microsoft Defender for Cloud.

Q37. How do you connect to a Linux VM in Azure?

✔ **Answer:** Connect to a Linux VM using SSH (Secure Shell) on port 22. You can use Azure Cloud Shell, terminal with ssh command, or PuTTY on Windows. Use SSH key pairs for authentication instead of passwords. Azure Bastion also supports SSH connections through the browser without exposing port 22 publicly.

Q38. What is Azure Bastion?

✔ **Answer:** Azure Bastion is a fully managed PaaS service that provides secure RDP/SSH access to VMs directly from the Azure portal over TLS — without needing a public IP on your VM. It eliminates exposure of management ports (22/3389) to the internet, reducing the attack surface significantly.

Q39. What is Just-In-Time (JIT) VM Access?

✔ **Answer:** JIT VM Access (via Microsoft Defender for Cloud) locks down inbound traffic to management ports (SSH, RDP) by default. When you need access, you request it for a specific duration (e.g., 3 hours), and the NSG rule is temporarily opened only for your IP. After the time expires, access is automatically revoked.

Q40. What is a VM Snapshot?

✔ **Answer:** A snapshot is a point-in-time read-only copy of a managed disk. Snapshots are useful for backups before making changes, creating new disks from snapshots, and migrating VMs. Snapshots are stored as standard storage and charged based on used space. They can be taken without stopping the VM (application-consistent recommended).

Q41. What is Azure VM Backup?

✔ **Answer:** Azure Backup provides VM-level backup using a Recovery Services Vault. It supports application-consistent backups for Windows (VSS) and Linux (pre/post scripts). You can set backup policies with daily/weekly/monthly retention. Backup data is stored in the vault and can be used to restore a full VM or individual disks/files.

Q42. What is Azure Site Recovery (ASR)?

✔ **Answer:** Azure Site Recovery provides disaster recovery by replicating Azure VMs (or on-premises machines) to a secondary Azure region. If the primary region fails, you can failover to the secondary. ASR provides RPO of minutes and RTO of hours. It also handles planned failover for maintenance and migration scenarios.

Q43. What is Custom Script Extension for Azure VMs?

✔ **Answer:** The Custom Script Extension allows you to run scripts on VMs after deployment — useful for post-deployment configuration like installing software, configuring settings, or running commands. Scripts can be stored in Azure Storage or GitHub. It runs once and is useful for automation without requiring agent setup.

Q44. What is Azure VM Serial Console?

✔ **Answer:** Azure Serial Console provides text-based console access to VMs for troubleshooting and diagnostics — even when the VM's network or RDP/SSH is not working. It connects to the VM's serial port and allows you to see boot messages, access Emergency Management Services (EMS) on Windows, or GRUB on Linux.

Q45. What is proximity placement group?

✔ **Answer:** Proximity Placement Groups (PPG) ensure that Azure VMs are physically located close together in the same datacenter to minimize network latency between them. This is critical for high-performance computing, clustered applications, and workloads that require ultra-low latency communication between VMs.

Q46. What is Azure Spot VMs?

✔ **Answer:** Spot VMs allow you to use Azure's unused capacity at up to 90% discount compared to pay-as-you-go prices. The trade-off is that Spot VMs can be evicted (stopped/deleted) by Azure with little notice when capacity is needed. They're suitable for fault-tolerant, batch processing, or interruptible workloads.

Q47. What are VM Extensions?

✔ **Answer:** VM Extensions are small applications that provide post-deployment configuration and automation tasks on Azure VMs. Examples include Custom Script Extension (run scripts), Dependency Agent (for VM Insights), Microsoft Monitoring Agent (Log Analytics), and Desired State Configuration (DSC) for configuration management.

Q48. What is Azure Hybrid Benefit for VMs?

✔ **Answer:** Azure Hybrid Benefit lets you use your existing on-premises Windows Server or SQL Server licenses with Software Assurance to run VMs on Azure at a reduced cost. This can save up to 49% on Windows VMs. Similarly, Linux Hybrid Benefit lets you bring Red Hat or SUSE subscriptions to Azure.

Q49. What is a Generalized vs Specialized VM image?

✔ **Answer:** A Generalized image (Sysprep for Windows, waagent deprovision for Linux) has machine-specific information removed and can be used to create multiple new VMs. A Specialized image is a copy of a specific VM retaining machine-specific info like hostname and user accounts — used to create an exact replica. Generalized is for templates; Specialized for clones.

Q50. What is Azure VM Image Builder?

✔ **Answer:** Azure VM Image Builder is a managed service based on HashiCorp Packer that automates the creation, validation, and distribution of custom VM images. You define an image template specifying the base image, customization scripts, and distribution targets. It creates standardized golden images for your organization.

3 Azure Networking

Q51. What is an Azure Virtual Network (VNet)?

✔ **Answer:** A VNet is the fundamental building block of private networking in Azure. It's a logically isolated network that you define, allowing Azure resources like VMs to securely communicate with each other, the internet, and on-premises networks. VNets are scoped to a region and can be segmented into subnets.

Q52. What is a Subnet in Azure?

✔ **Answer:** A subnet is a segment of a VNet's IP address space. You divide a VNet into subnets to organize resources and apply security controls. Each resource deployed in Azure (like a VM) goes into a subnet. You can have separate subnets for web tier, app tier, and database tier to isolate traffic.

Q53. What is VNet Peering?

✔ **Answer:** VNet Peering connects two Azure VNets, allowing resources in different VNets to communicate using private IP addresses as if they were in the same network. Traffic routes through the Microsoft backbone — not the internet — making it fast and private. VNet Peering can be within the same region (regional) or across regions (global peering).

Q54. What is an NSG (Network Security Group)?

✔ **Answer:** An NSG is a security filter that controls inbound and outbound network traffic to Azure resources. It contains security rules with priority, source/destination, port, and protocol. NSGs can be applied to subnets or individual NICs. Rules are evaluated by priority — lower number = higher priority. Default rules allow VNet-to-VNet and deny all internet inbound.

Q55. What is Azure Load Balancer?

✔ **Answer:** Azure Load Balancer is a Layer 4 (TCP/UDP) load balancer that distributes incoming traffic across backend VMs. Public Load Balancer handles internet-facing traffic; Internal Load Balancer handles internal traffic within a VNet. It supports health probes to automatically remove unhealthy VMs. Provides 99.99% SLA.

Q56. What is Azure Application Gateway?

✔ **Answer:** Application Gateway is a Layer 7 (HTTP/HTTPS) load balancer and web application firewall (WAF). It can route traffic based on URL path or host header, perform SSL termination, provide cookie-based session affinity, and protect against OWASP web vulnerabilities. It's suitable for web applications requiring advanced routing.

Q57. What is Azure Traffic Manager?

✔ **Answer:** Traffic Manager is a DNS-based global load balancer that routes users to the nearest or healthiest endpoint across multiple regions. Routing methods include Performance (lowest latency), Weighted (traffic split), Priority (failover), Geographic (location-based), and Multivalue. It operates at DNS level — not packet level.

Q58. What is Azure Front Door?

✔ **Answer:** Azure Front Door is a global HTTP load balancer and CDN with built-in WAF capabilities. It routes HTTP/HTTPS traffic to the nearest available backend, provides SSL offload, URL-based routing, caching, and DDoS protection. It's designed for globally distributed web applications needing low latency and high availability.

Q59. What is Azure DNS?

✔ **Answer:** Azure DNS hosts DNS domains and provides name resolution using Microsoft's global DNS infrastructure. It's highly available and fast. You can host public DNS zones (for internet-facing names) and private DNS zones (for name resolution within VNets without a custom DNS server). Azure DNS is integrated with Azure RBAC.

Q60. What is a Private DNS Zone in Azure?

✔ **Answer:** Private DNS Zones provide DNS name resolution within Azure VNets without custom DNS servers. They're linked to VNets for auto-registration of VM hostnames or manual record creation. Used with Private Endpoints so that services like Azure Storage resolve to private IPs within your VNet instead of public IPs.

Q61. What is Azure VPN Gateway?

✔ **Answer:** VPN Gateway is a managed gateway that sends encrypted traffic between Azure VNets and on-premises networks over the internet using IPsec/IKE tunnels. Supports Site-to-Site VPN (always-on, connects entire network), Point-to-Site VPN (per-user VPN for remote workers), and VNet-to-VNet connections.

Q62. What is Azure ExpressRoute?

✔ **Answer:** ExpressRoute provides a private, dedicated network connection between your on-premises infrastructure and Azure — bypassing the public internet. It's delivered through connectivity providers and offers higher reliability, faster speeds (50 Mbps to 100 Gbps), lower latency, and higher security than VPN. Ideal for enterprise workloads with strict network requirements.

Q63. What is the difference between VPN Gateway and ExpressRoute?

✔ **Answer:** VPN Gateway uses the public internet with IPsec encryption — lower cost, easier setup, but variable latency and throughput. ExpressRoute is a private dedicated connection through a provider — higher cost, requires provider setup, but offers guaranteed bandwidth, lower latency, and higher security. Choose ExpressRoute for critical enterprise connectivity.

Q64. What is Azure Private Link?

✔ **Answer:** Private Link allows you to access Azure PaaS services (Storage, SQL, Key Vault, etc.) over a private endpoint in your VNet. Instead of connecting over the internet, traffic stays on the Microsoft backbone. This eliminates data exfiltration risks, as services get a private IP within your VNet.

Q65. What is a Private Endpoint?

✔ **Answer:** A Private Endpoint is a network interface in your VNet that connects you privately to an Azure service powered by Azure Private Link. The service gets a private IP in your subnet. DNS configuration ensures the service's FQDN resolves to this private IP, so all traffic stays within Azure's private network.

Q66. What is a Service Endpoint?

✔ **Answer:** Service Endpoints extend your VNet identity to Azure services (Storage, SQL, Key Vault) over optimized routes on the Azure backbone. Unlike Private Endpoints, service endpoints don't assign a private IP — the service still has a public IP but access is restricted to your VNet. Service Endpoints are simpler but less secure than Private Endpoints.

Q67. What is Azure Firewall?

✔ **Answer:** Azure Firewall is a managed, cloud-native network security service that protects Azure VNet resources. It provides stateful packet inspection, threat intelligence-based filtering, FQDN filtering (allow/deny by domain name), DNAT rules for inbound NAT, network rules, and application rules. It scales automatically with your traffic.

Q68. What is Azure DDoS Protection?

✔ **Answer:** Azure DDoS Protection defends against Distributed Denial of Service attacks. The Basic tier is free and automatically enabled for all Azure resources. The Standard tier (DDoS Protection Plan) provides advanced protection for resources in VNets, including adaptive tuning, attack analytics, rapid response support, and cost protection.

Q69. What is Azure Bastion and why is it important for networking?

✔ **Answer:** Azure Bastion is a fully managed jump host deployed in your VNet that provides secure RDP/SSH access to VMs over the Azure portal using TLS — without requiring public IPs or opening RDP/SSH ports. This eliminates the need for a traditional jump box VM and reduces the attack surface of your Azure environment.

Q70. What is Network Watcher in Azure?

✔ **Answer:** Azure Network Watcher is a regional service providing monitoring, diagnostics, and logging for Azure network resources. Key features include: Connection Monitor (test connectivity between VMs), NSG Flow Logs (log all traffic through NSGs), IP Flow Verify (test if traffic is allowed/denied), Packet Capture, and VPN diagnostics.

Q71. What is NSG Flow Logs?

✔ **Answer:** NSG Flow Logs capture information about IP traffic flowing through an NSG. Each log entry includes 5-tuple info (source/destination IP, source/destination port, protocol) and whether traffic was allowed or denied. Logs are stored in Azure Storage and can be analyzed with Network Watcher Traffic Analytics or SIEM tools.

Q72. What is User Defined Routes (UDR)?

✔ **Answer:** UDR (also called custom routes or route tables) allow you to override Azure's default routing and control where network traffic is directed. For example, you can route all outbound internet traffic through Azure Firewall or a network virtual appliance for inspection. You associate a route table with a subnet.

Q73. What is Azure NAT Gateway?

✔ **Answer:** Azure NAT Gateway provides outbound internet connectivity for resources in a subnet using static public IP addresses. All VMs in the subnet share the NAT gateway's IP(s) for outbound traffic, providing SNAT port exhaustion prevention and a predictable outbound IP — important for allowlisting on external firewalls.

Q74. What is the difference between Public and Private IP in Azure?

✔ **Answer:** Public IPs are routable over the internet and assigned to resources like VMs, Load Balancers, and Application Gateways. Private IPs are from your VNet address space and only routable within the VNet and connected networks. Most backend resources use only private IPs, with public IPs reserved for internet-facing resources.

Q75. What is a Network Interface Card (NIC) in Azure VMs?

✔ **Answer:** A NIC is the connection point between a VM and a VNet. Each VM has at least one NIC with a private IP (and optionally a public IP). Some VM sizes support multiple NICs for traffic separation (e.g., management vs. data). The NIC also has NSG that can be applied to it for fine-grained traffic control.

Q76. What is Azure Virtual WAN?

✔ **Answer:** Azure Virtual WAN is a networking service that brings networking, security, and routing functionalities together into a single operational interface. It provides optimized branch-to-branch and branch-to-Azure connectivity via a hub-and-spoke architecture. It integrates with SD-WAN and VPN partners for automated setup.

Q77. What is a VNet Gateway Subnet?

✔ **Answer:** The GatewaySubnet is a dedicated subnet required when deploying a VPN Gateway or ExpressRoute Gateway in a VNet. It must be named 'GatewaySubnet' exactly. It should be at least /27 in size and should not contain any other resources — only the gateway instances go here.

Q78. What is connection troubleshooting in Azure Network Watcher?

✔ **Answer:** Connection Troubleshoot is a Network Watcher feature that checks end-to-end connectivity between a source VM and a destination (VM, FQDN, or IP:port). It identifies network issues like NSG blocks, routing problems, or VM-level issues. It's a quick way to diagnose 'can VM A reach VM B' questions.

Q79. What is Azure CDN (Content Delivery Network)?

✔ **Answer:** Azure CDN delivers static content (images, JS, CSS, videos) from edge servers closest to users worldwide, reducing latency. Content is cached at edge locations based on TTL rules. Azure CDN integrates with Azure Storage, Web Apps, and custom origins. It supports HTTPS, custom domains, geo-filtering, and compression.

Q80. What is the difference between Azure Load Balancer SKUs (Basic vs Standard)?

✔ **Answer:** Standard SKU supports Availability Zones, HTTPS health probes, diagnostic logging, higher port rules, and outbound rules. Basic SKU is free but limited — no zone support, no HTTPS probes, no outbound rules. Standard is recommended for production. VMs in Standard Load Balancer back-end pools need Standard public IPs.

4 Azure Storage

Q81. What are the main Azure Storage services?

✔ **Answer:** Azure Storage includes four core services: Blob Storage (object storage for unstructured data like files, images, videos), File Storage (managed file shares via SMB/NFS), Queue Storage (message queuing for decoupled apps), and Table Storage (NoSQL key-value store). All are accessed through Azure Storage Accounts.

Q82. What is an Azure Storage Account?

✔ **Answer:** A Storage Account is a unique namespace in Azure for your storage data. All storage objects (blobs, files, queues, tables) are accessible via the storage account endpoint. It provides high availability, durability, scalability, and security. Access control is managed via access keys, SAS tokens, or Azure AD.

Q83. What are the Storage Account types in Azure?

✔ **Answer:** General Purpose v2 (GPv2) supports all storage services and features — recommended for most use cases. General Purpose v1 (GPv1) is older with fewer features. Blob Storage accounts are for Blob-only workloads. BlockBlobStorage and FileStorage are premium performance accounts for specific use cases.

Q84. What are the Blob Storage access tiers?

✔ **Answer:** Hot tier: frequent access, higher storage cost, lower access cost. Cool tier: infrequent access (at least 30 days), lower storage cost, higher access cost. Cold tier: rarely accessed data (at least 90 days). Archive tier: rarely accessed, lowest cost, but data is offline and must be rehydrated (hours) before access.

Q85. What is Blob Storage rehydration?

✔ **Answer:** Rehydration is the process of changing an archived blob back to Hot or Cool tier so it can be accessed. This can take up to 15 hours for standard priority or up to 1 hour for high priority (with additional cost). You can copy an archived blob to a new blob in Hot/Cool tier without affecting the original.

Q86. What is Azure Blob Storage lifecycle management?

✔ **Answer:** Lifecycle management allows you to define rules that automatically transition blobs between tiers (Hot → Cool → Archive) or delete blobs based on age or last modified date. For example: move to Cool after 30 days, Archive after 90 days, delete after 365 days. This optimizes storage costs automatically.

Q87. What is the difference between Block Blob, Page Blob, and Append Blob?

✔ **Answer:** Block Blobs store text and binary data in blocks — ideal for documents, images, and videos. Append Blobs are optimized for append operations (like log files) where data is added to the end. Page Blobs store random-access files up to 8 TB — used as VHD files (Azure VM disks). Most use cases use Block Blobs.

Q88. What is Azure File Storage?

✔ **Answer:** Azure File Storage provides fully managed file shares in the cloud accessible via SMB 3.0 and NFS 4.1 protocols. It can replace on-premises file servers. Azure Files supports Windows, Linux, and macOS. Files can be mounted directly on VMs or sync'd using Azure File Sync for hybrid scenarios.

Q89. What is Azure File Sync?

✔ **Answer:** Azure File Sync transforms Windows Server into a quick cache of your Azure file shares. It syncs files between on-premises file servers and Azure Files, enabling cloud tiering (less-used files stored in cloud, frequently-used cached locally). This extends Azure Files to on-premises environments seamlessly.

Q90. What is a Shared Access Signature (SAS)?

✔ **Answer:** A SAS is a URI that grants restricted access to Azure Storage resources for a specific time period. You can specify which services, resource types, permissions (read/write/delete), start/expiry times, and allowed IPs. There are three types: Account SAS, Service SAS, and User Delegation SAS (uses Azure AD credentials — most secure).

Q91. What is Azure Storage Encryption?

✔ **Answer:** All data in Azure Storage is encrypted at rest using 256-bit AES encryption (SSE — Storage Service Encryption) automatically. By default, Microsoft manages encryption keys. You can use Customer-Managed Keys (CMK) stored in Azure Key Vault for more control. Encryption is transparent and doesn't impact performance.

Q92. What is Azure Storage replication?

✔ **Answer:** Azure Storage offers multiple replication options: LRS (Locally Redundant Storage, 3 copies in one datacenter), ZRS (Zone-Redundant Storage, across 3 zones in a region), GRS (Geo-Redundant Storage, LRS + async copy to secondary region), GZRS (ZRS + async copy to secondary region). Higher redundancy = higher cost + better durability.

Q93. What is the difference between GRS and RA-GRS?

✔ **Answer:** GRS replicates to a secondary region but the secondary is read-only ONLY during failover. RA-GRS (Read-Access Geo-Redundant Storage) allows read access to secondary region data at all times (using -secondary endpoint), even when primary is healthy. RA-GRS is useful for read workloads that can tolerate eventual consistency.

Q94. What is Azure Data Lake Storage Gen2?

✔ **Answer:** ADLS Gen2 combines Azure Blob Storage with a hierarchical namespace (like a file system) for big data analytics. It's built on top of Azure Blob Storage, so it supports all blob features. It integrates natively with Azure Databricks, HDInsight, and Synapse Analytics. It uses POSIX-style permissions for fine-grained access control.

Q95. What is Azure Queue Storage?

✔ **Answer:** Queue Storage stores large numbers of messages for asynchronous communication between application components. Messages can be up to 64 KB in size. Applications read messages from queues for processing. This decoupling enables scalable, resilient architectures — if a component fails, messages stay in queue until processed.

Q96. What is Azure Table Storage?

✔ **Answer:** Table Storage is a NoSQL key-value store for storing large amounts of structured, non-relational data. Data is organized in tables with rows (entities) containing up to 252 custom properties. It's accessed via OData protocol. For more advanced features, Azure Cosmos DB Table API is recommended as it's fully compatible.

Q97. What is Immutable Blob Storage?

✔ **Answer:** Immutable storage allows you to store blobs in a WORM (Write Once, Read Many) state. Once data is written, it can't be modified or deleted for a specified retention period. This is used for compliance with regulations like SEC 17a-4, FINRA, etc. You can configure time-based retention policies or legal holds.

Q98. What is Soft Delete in Azure Storage?

✔ **Answer:** Soft Delete protects blobs, containers, and file shares from accidental deletion. When enabled, deleted items are retained for a configurable period (1-365 days) and can be recovered. This is a safety net against accidental deletes or ransomware attacks. After the retention period, items are permanently deleted.

Q99. What is Azure Storage Explorer?

✔ **Answer:** Azure Storage Explorer is a free, standalone desktop application from Microsoft that makes it easy to work with Azure Storage data on Windows, macOS, and Linux. You can upload, download, and manage blobs, files, queues, tables, and Cosmos DB entities. It supports Azure AD authentication and SAS tokens.

Q100. What is object replication in Azure Blob Storage?

✔ **Answer:** Object Replication asynchronously copies block blobs between storage accounts (same or different regions). You configure replication rules specifying source and destination containers, filter by prefix/tag, and optionally copy only new blobs. Unlike GRS (which replicates everything), object replication gives granular control.

Q101. What is Azure Storage Firewall?

✔ **Answer:** Storage Firewall allows you to restrict access to your storage account by configuring allowed IP ranges, VNet service endpoints, or Private Endpoints. By default, storage accounts accept connections from any network. Enabling firewall rules means only specified networks/IPs can access the storage account.

Q102. What is Azure Blob Storage versioning?

✔ **Answer:** Blob versioning automatically maintains previous versions of a blob when it's modified or deleted. Each version has a version ID. You can restore blobs to previous versions — useful for data protection and recovery from accidental overwrites. Versioning has storage cost implications as all versions are retained.

Q103. What is Azure Premium Storage?

✔ **Answer:** Premium Storage uses SSDs to provide high-performance, low-latency storage for VMs. Premium SSD supports up to 20,000 IOPS and 900 MB/s throughput per disk. Premium SSD v2 supports up to 80,000 IOPS. Ultra Disk supports up to 160,000 IOPS with sub-millisecond latency for the most demanding workloads.

Q104. What is the Storage Account key and how is it secured?

✔ **Answer:** Each storage account has two 512-bit access keys that grant full control over the account. They should be treated like passwords. Best practice: rotate keys regularly, store in Azure Key Vault, use SAS tokens or Azure AD instead of keys for app access. Microsoft Defender for Storage alerts on suspicious activity.

Q105. What is static website hosting in Azure Blob Storage?

✔ **Answer:** Azure Blob Storage supports hosting static websites (HTML, CSS, JS, images) directly from a $web container. You get a website endpoint URL. No server required — ideal for static sites, SPAs, and documentation sites. For custom domains with HTTPS and CDN features, pair with Azure Front Door or Azure CDN.

5 Azure Monitor & Logging

Q106. What is Azure Monitor?

✔ **Answer:** Azure Monitor is a comprehensive monitoring service that collects, analyzes, and acts on telemetry from Azure resources and on-premises environments. It includes features like Metrics (numerical time-series data), Logs (structured/unstructured log data in Log Analytics), Alerts, Dashboards, and Workbooks — providing end-to-end observability.

Q107. What is the difference between Azure Monitor Metrics and Logs?

✔ **Answer:** Metrics are lightweight numerical time-series data (CPU%, network bytes) stored for 93 days, optimized for real-time alerting and dashboards. Logs (in Log Analytics) store structured/unstructured text data from various sources, queryable with Kusto Query Language (KQL). Logs provide deeper diagnostics but require more query effort.

Q108. What is Log Analytics Workspace?

✔ **Answer:** A Log Analytics Workspace is a unique environment for Azure Monitor log data. Resources send their log/metric data to the workspace. You query data using KQL (Kusto Query Language). Multiple resources (VMs, App Services, Key Vault) can send data to the same workspace, enabling cross-resource queries.

Q109. What is KQL (Kusto Query Language)?

✔ **Answer:** KQL is the query language used to analyze data in Azure Log Analytics, Azure Data Explorer, and Microsoft Sentinel. It uses a pipe (|) syntax: start with a table, then filter, summarize, join, project, etc. Example: AzureActivity | where OperationName == 'Start Virtual Machine' | summarize count() by ResourceGroup.

Q110. What is Azure Activity Log?

✔ **Answer:** The Activity Log records subscription-level events — who did what and when in your Azure subscription. It captures management operations like creating/deleting resources, changing policies, and role assignments. Activity logs are retained for 90 days by default. They can be sent to Log Analytics for longer retention and querying.

Q111. What is Azure Diagnostic Settings?

✔ **Answer:** Diagnostic Settings configure where a resource sends its platform logs and metrics. You can route logs to Log Analytics Workspace (for querying), Storage Account (for archival), Event Hub (for external SIEM), or Partner solutions. Not all resources emit the same log categories — check each resource's documentation.

Q112. What is Azure Alerts?

✔ **Answer:** Azure Alerts notify you when specific conditions are met in your monitoring data. You can create alerts based on Metrics (numerical thresholds), Log queries (KQL result conditions), Activity Log events, and Service Health. Alerts can trigger Action Groups to send emails, SMS, call webhooks, run Logic Apps, or create ITSM tickets.

Q113. What is an Action Group?

✔ **Answer:** An Action Group defines what happens when an alert fires — it's a collection of notification and action types. Notifications include Email, SMS, Push (Azure app), Voice. Actions include Webhook, Logic App, Azure Function, ITSM connector, Runbook, and Event Hub. Multiple alerts can share the same action group.

Q114. What is Azure Application Insights?

✔ **Answer:** Application Insights is an APM (Application Performance Monitoring) service for live web applications. It automatically detects performance anomalies and includes analytics tools to diagnose issues. It monitors requests, dependencies, exceptions, page views, and user behavior. It's built on top of Azure Monitor (data stored in Log Analytics).

Q115. What is Azure Service Health?

✔ **Answer:** Azure Service Health tracks the health of Azure services and regions, informing you about incidents that may affect your resources. It includes Service issues (ongoing outages), Planned maintenance, and Health advisories (degraded but not affecting you yet). You can create alerts for specific service/region combinations.

Q116. What is Azure Resource Health?

✔ **Answer:** Resource Health shows the current and historical health of your specific Azure resources — not the platform in general. It tells you if your VM, App Service, or SQL Database is Available, Unavailable, Degraded, or Unknown, and the reason. It helps distinguish between Azure platform issues and issues in your own resource configuration.

Q117. What are Azure Monitor Workbooks?

✔ **Answer:** Workbooks provide a flexible canvas for data analysis and visualization in Azure Monitor. They combine text, KQL queries, metrics, and parameters into rich interactive reports. Pre-built workbook templates exist for common scenarios like VM performance, network monitoring, and security compliance. They're shareable within the Azure portal.

Q118. What is Azure Monitor Dashboards?

✔ **Answer:** Azure Dashboards are customizable UI pages in the Azure portal that pin tiles from various Azure resources — metrics charts, log query results, resource lists, and alerts. They provide a single pane of glass for monitoring. Dashboards can be shared with others in your organization through RBAC.

Q119. What is the difference between Azure Monitor and Microsoft Sentinel?

✔ **Answer:** Azure Monitor focuses on infrastructure and application performance monitoring. Microsoft Sentinel is a cloud-native SIEM (Security Information and Event Management) and SOAR (Security Orchestration and Automated Response) solution that focuses on security analytics, threat detection, investigation, and response using data from Monitor and other sources.

Q120. What is Metric Alerts vs Log Alerts in Azure Monitor?

✔ **Answer:** Metric Alerts evaluate numerical metric values at regular intervals and fire when a threshold is crossed (e.g., CPU > 90%). They're stateful — fire when condition is met, resolve when cleared. Log Alerts run KQL queries on a schedule and alert based on query results. Log Alerts are more flexible but less real-time than metric alerts.

Q121. What is Azure Monitor Agent (AMA)?

✔ **Answer:** Azure Monitor Agent is the latest generation unified agent that replaces Log Analytics Agent (MMA/OMS) and Diagnostics Extension. It collects performance data, event logs, and syslog from VMs and sends to Log Analytics Workspaces or Azure Monitor Metrics. It uses Data Collection Rules (DCR) for flexible data routing.

Q122. What is VM Insights in Azure Monitor?

✔ **Answer:** VM Insights is a feature of Azure Monitor that provides pre-built monitoring for VMs — including performance charts for CPU, memory, disk, and network, along with a map of dependencies showing which processes are running and how they connect to other services. It requires Azure Monitor Agent and Dependency Agent.

Q123. What is Container Insights?

✔ **Answer:** Container Insights is Azure Monitor's solution for monitoring containerized workloads on AKS and Azure Arc-enabled Kubernetes clusters. It collects metrics and logs from containers, pods, and nodes, providing pre-built dashboards for CPU/memory utilization, pod failures, and container logs.

Q124. What is a Data Collection Rule (DCR)?

✔ **Answer:** DCRs define what data to collect from resources and where to send it. They allow you to filter events, select specific performance counters, and route data to different destinations (multiple Log Analytics workspaces). DCRs replace legacy agent configuration and provide centralized, scalable data collection management.

Q125. What is Azure Monitor Private Link Scope (AMPLS)?

✔ **Answer:** AMPLS allows Azure Monitor resources (Log Analytics workspaces, Application Insights) to be accessed over Private Endpoints, ensuring monitoring traffic stays within your private network and doesn't traverse the internet. This is important for organizations with strict network isolation requirements.

6 Azure Resource Management

Q126. What is Azure Resource Manager (ARM)?

✔ **Answer:** ARM is the deployment and management service for Azure. It provides a management layer to create, update, and delete Azure resources. All operations (portal, CLI, PowerShell, APIs, Terraform) go through ARM. ARM handles authentication via Azure AD, provides RBAC, resource locking, tagging, and consistent API across all tools.

Q127. What is an ARM Template?

✔ **Answer:** ARM Templates are JSON files that define Azure infrastructure as code (IaC). They describe resources, their configurations, and dependencies in a declarative way — you specify the desired state and ARM figures out how to achieve it. Templates enable repeatable, consistent deployments. They support parameters, variables, functions, and outputs.

Q128. What is Bicep?

✔ **Answer:** Bicep is a domain-specific language (DSL) for deploying Azure resources that compiles to ARM JSON. It offers cleaner syntax, better readability, type safety, and IntelliSense support in VS Code. Bicep is Microsoft's recommended approach for new IaC deployments on Azure, replacing raw ARM JSON templates.

Q129. What is a Resource Group in Azure?

✔ **Answer:** A Resource Group is a logical container that holds related Azure resources for a solution. All resources in a group share the same lifecycle — created, managed, and deleted together. Resources in a group don't need to be in the same region. RBAC and policies applied to a group affect all resources in it.

Q130. What are Azure Subscriptions?

✔ **Answer:** A Subscription is a billing and access management unit in Azure. Resources are created within subscriptions, and costs are billed per subscription. You can have multiple subscriptions to separate environments (Dev, Test, Prod), departments, or projects. Each subscription has limits (quotas) on the number of resources you can create.

Q131. What is Azure Management Groups?

✔ **Answer:** Management Groups let you organize subscriptions into a hierarchy for applying governance at scale. You can apply Azure Policy, RBAC, and compliance requirements to a management group, and all subscriptions underneath inherit them. The Root Management Group is at the top, and you can create up to 6 levels of nesting.

Q132. What is Azure Policy?

✔ **Answer:** Azure Policy enforces organizational standards and assesses compliance across your Azure resources. You define policies (rules) that resources must comply with. Policies can Audit (report non-compliance), Deny (block non-compliant deployments), or DeployIfNotExists (auto-remediate). Policy is assigned at management group, subscription, or resource group scope.

Q133. What is a Policy Initiative (Policy Set Definition)?

✔ **Answer:** A Policy Initiative is a collection of policy definitions grouped together to achieve a broader compliance goal. For example, the 'CIS Microsoft Azure Foundations Benchmark' initiative contains dozens of individual policies. Using initiatives simplifies managing and assigning related policies as a single unit.

Q134. What is Azure Blueprints?

✔ **Answer:** Azure Blueprints allow you to define a repeatable set of Azure resources including ARM templates, policies, RBAC assignments, and resource groups — and deploy them together in one operation. Blueprints track which resources were deployed from which blueprint version, maintaining traceability. Useful for regulated environments needing consistent setups.

Q135. What is Resource Locking in Azure?

✔ **Answer:** Resource Locks prevent accidental deletion or modification of critical resources. CanNotDelete lock allows read and modify but blocks delete. ReadOnly lock allows read but blocks modify and delete. Locks can be applied at subscription, resource group, or individual resource level. Admin access doesn't bypass locks.

Q136. What are Azure Tags?

✔ **Answer:** Tags are name-value pairs you apply to Azure resources for organization and cost management. Example: Environment=Production, Team=Finance, CostCenter=12345. You can filter resources, generate cost reports, and apply governance policies based on tags. Tags are not inherited — you must apply them explicitly to each resource (or use Policy to enforce tagging).

Q137. What is the difference between Azure Policy and RBAC?

✔ **Answer:** RBAC controls WHO can do what (access control) — it defines which users/groups can perform which actions. Azure Policy controls WHAT can be done (governance) — it enforces rules about what resources can be created or how they must be configured. Both are complementary and often used together for complete governance.

Q138. What is a Deployment Stack in Azure?

✔ **Answer:** Deployment Stacks (newer feature) allow you to manage a collection of Azure resources as a single atomic unit using Bicep or ARM templates. Unlike resource groups, a stack can deny modifications/deletions to managed resources (with deny settings). It provides lifecycle management and rollback capabilities for template-deployed resources.

Q139. What is the Azure Cloud Shell?

✔ **Answer:** Azure Cloud Shell is a browser-based, authenticated shell experience for managing Azure resources — accessible from the Azure portal, shell.azure.com, or VS Code extension. It provides both Bash and PowerShell environments with Azure CLI, Azure PowerShell, Terraform, Ansible, and other tools pre-installed. Files persist in an Azure Storage share.

Q140. What is Azure CLI?

✔ **Answer:** Azure CLI is a cross-platform command-line tool for managing Azure resources from Windows, macOS, and Linux. Commands follow az syntax (e.g., az vm create, az group list). It's scriptable for automation. Authentication uses az login. Azure CLI is great for DevOps pipelines and automation scripts.

Q141. What is Azure PowerShell?

✔ **Answer:** Azure PowerShell is a set of cmdlets for managing Azure resources using PowerShell. Commands follow Verb-AzNoun patterns (e.g., New-AzVM, Get-AzResourceGroup). It integrates with PowerShell scripting and automation. Both Azure CLI and Azure PowerShell can accomplish the same tasks — choice depends on team preference.

Q142. What is the Azure Portal?

✔ **Answer:** The Azure Portal (portal.azure.com) is a web-based, unified management console for managing Azure resources via a graphical interface. You can create, configure, monitor, and manage all Azure services from the portal. It provides dashboards, wizards, and shortcuts for common tasks. It also has a mobile app.

Q143. What is Azure Cost Management + Billing?

✔ **Answer:** Azure Cost Management provides tools to monitor, allocate, and optimize Azure spending. You can view cost breakdowns by subscription, resource group, resource, or tag. Create budgets with alert notifications when spending thresholds are crossed. Cost analysis charts show spending trends. Recommendations identify idle or right-sizing opportunities.

Q144. What are Azure Reservations?

✔ **Answer:** Azure Reservations allow you to commit to a 1-year or 3-year plan for VMs, SQL Databases, Azure Cosmos DB, and other services in exchange for significant discounts (up to 72% compared to pay-as-you-go). You pay upfront or monthly. Reservations are best for predictable, steady-state workloads.

Q145. What is the Azure Pricing Calculator?

✔ **Answer:** The Azure Pricing Calculator (azure.microsoft.com/pricing/calculator) helps you estimate the monthly cost of Azure services before you deploy them. You configure services (VM size, storage, region, redundancy) and get an estimated cost. This helps with budgeting and comparing different architecture options.

Q146. What is Azure Advisor?

✔ **Answer:** Azure Advisor is a personalized cloud consultant that analyzes your Azure configurations and usage to provide recommendations across five categories: Reliability (high availability), Security (vulnerability fixes), Performance (speed improvements), Cost (spending optimization), and Operational Excellence (best practices).

Q147. What is a Naming Convention in Azure?

✔ **Answer:** Azure naming conventions use a consistent pattern for resource names — including resource type, workload, environment, region, and instance number. Example: vm-myapp-prod-eastus-001. Clear naming makes resources identifiable, reduces confusion, and enables filtering/scripting. Microsoft provides naming convention guidance in the Cloud Adoption Framework.

Q148. What is Azure Landing Zone?

✔ **Answer:** An Azure Landing Zone is a pre-configured, secure, and scalable Azure environment built following best practices from the Cloud Adoption Framework. It includes management groups hierarchy, policies, RBAC, networking, and logging set up consistently. It's the starting point for enterprise Azure adoption at scale.

Q149. What is the Cloud Adoption Framework (CAF)?

✔ **Answer:** CAF is Microsoft's collection of documentation, best practices, and tools to help organizations adopt Azure effectively. It covers strategy, planning, readiness, adoption, governance, and management. It provides prescriptive guidance from initial business case to optimized running workloads, including Landing Zone designs.

Q150. What is Azure Managed Applications?

✔ **Answer:** Azure Managed Applications allow service providers to build and offer pre-configured Azure solutions where the publisher maintains the managed resource group. Customers deploy the solution without needing deep Azure expertise, while the publisher retains management control. Published via Azure Marketplace or internal service catalogs.

7 Azure App Services & PaaS

Q151. What is Azure App Service?

✔ **Answer:** Azure App Service is a fully managed PaaS platform for hosting web applications, RESTful APIs, and mobile backends. Supports multiple languages: .NET, Java, Node.js, Python, PHP, Ruby. You don't manage the underlying infrastructure — Azure handles OS patching, scaling, load balancing, and security. Available on Windows and Linux.

Q152. What is an App Service Plan?

✔ **Answer:** An App Service Plan defines the compute resources (CPU, RAM, disk, OS) for your App Service apps. All apps in the same plan share its resources. Pricing tiers: Free (dev/test), Shared, Basic (no auto-scale), Standard (auto-scale, custom domains), Premium (VNet integration, more performance), Isolated (dedicated environment).

Q153. What is Auto-Scaling in App Service?

✔ **Answer:** Auto-scaling automatically adds or removes App Service Plan instances based on defined rules. Scale-out triggers: CPU > 80%, HTTP queue length, custom metrics, schedules. This ensures your app handles increased load without manual intervention. Standard tier and above support auto-scale (Basic supports manual scale only).

Q154. What is Deployment Slots in App Service?

✔ **Answer:** Deployment Slots are live app environments with their own URLs (e.g., myapp-staging.azurewebsites.net). You deploy to a staging slot, test it, then 'swap' it to production — achieving zero-downtime deployments. The swap warms up the new version and instantly switches traffic. Standard tier and above support slots.

Q155. What is Azure App Service VNet Integration?

✔ **Answer:** VNet Integration allows App Service apps to make outbound calls to resources in a VNet (like databases, internal APIs) using private IPs. It doesn't let the VNet initiate connections into the app — for inbound private access, use Private Endpoints. Requires Premium tier for regional VNet integration.

Q156. What is Azure App Service Environment (ASE)?

✔ **Answer:** ASE is a fully isolated, dedicated deployment of App Service into a customer's VNet. It provides the highest scale, complete network isolation, and maximum security. ASE supports internal (private) or external (internet-facing) load balancers. Suitable for enterprise apps needing strict compliance and network isolation.

Q157. What is Azure Functions?

✔ **Answer:** Azure Functions is a serverless compute service that runs code in response to events without managing infrastructure. You pay only for execution time and resources consumed (Consumption plan). Supports triggers: HTTP, Timer, Blob Storage, Service Bus, Event Hub, Cosmos DB, etc. Ideal for event-driven microservices and automation.

Q158. What is the difference between Azure Functions Consumption and Premium plans?

✔ **Answer:** Consumption plan scales automatically, bills per execution, but has a cold start (delay on first invocation after idle). Premium plan keeps instances pre-warmed (no cold start), supports VNet integration, unlimited execution duration, and larger instances. Dedicated (App Service) plan runs Functions on always-on App Service Plan instances.

Q159. What is Azure Logic Apps?

✔ **Answer:** Logic Apps is a cloud-based workflow automation service for integrating apps, data, services, and systems. You create workflows visually using 400+ connectors (Office 365, Salesforce, Twitter, SQL, SAP). It handles B2B messaging, enterprise integration, and automation without writing much code. Similar to Microsoft Power Automate.

Q160. What is Azure API Management (APIM)?

✔ **Answer:** APIM is a fully managed service for publishing, securing, and analyzing APIs. It acts as a proxy for your backend APIs, adding features like authentication, rate limiting, caching, transformation, and analytics — without changing the backend. It provides a developer portal for API documentation and subscription management.

Q161. What is Azure Container Instances (ACI)?

✔ **Answer:** ACI is the fastest and simplest way to run containers in Azure without managing VMs or orchestrators. You run containers on-demand (per-second billing) with no infrastructure to manage. Good for isolated tasks, burst computing, event-driven processing, and CI/CD pipelines. For production microservices, AKS is recommended.

Q162. What is Azure Kubernetes Service (AKS)?

✔ **Answer:** AKS is a managed Kubernetes service that simplifies deploying, managing, and scaling containerized applications using Kubernetes. Azure manages the Kubernetes control plane (API server, etcd, scheduler) for free — you only pay for the worker nodes. AKS integrates with Azure AD, Azure Monitor, ACR, and Azure networking.

Q163. What is Azure Container Registry (ACR)?

✔ **Answer:** ACR is a managed, private Docker container registry built on Docker Registry 2.0. Store, manage, and deploy container images securely. ACR Tasks automate image builds on code commits. Geo-replication makes images available in multiple regions. AKS and ACI integrate natively with ACR for image pulls.

Q164. What is Azure Static Web Apps?

✔ **Answer:** Azure Static Web Apps is a service that automatically builds and deploys full-stack web apps from GitHub or Azure DevOps. It serves static assets from a CDN globally, provides serverless API via Azure Functions, handles routing, and includes auth via Azure AD, GitHub, Twitter, etc. Free tier available for personal projects.

Q165. What is Azure App Configuration?

✔ **Answer:** App Configuration is a managed service for centrally managing application settings and feature flags. Instead of storing config in app code or multiple config files, all settings are stored centrally and accessed at runtime. It supports hierarchical keys, labels (for different environments), and integrates with Key Vault for secrets.

Q166. What is Azure Cache for Redis?

✔ **Answer:** Azure Cache for Redis is a managed Redis cache service for high-performance, low-latency data access. Used for caching frequently accessed data, session state, message queuing, and pub/sub scenarios. Redis stores data in memory for microsecond response times. Azure manages patching, monitoring, and high availability.

Q167. What is Azure Service Bus?

✔ **Answer:** Service Bus is a fully managed enterprise message broker with message queues and publish/subscribe topics. Unlike Queue Storage, Service Bus supports advanced messaging features: ordered delivery, duplicate detection, dead-lettering, sessions, transactions, and long message retention. Ideal for decoupling microservices and enterprise integration.

Q168. What is Azure Event Hub?

✔ **Answer:** Event Hub is a big data streaming platform capable of ingesting millions of events per second. It's used for telemetry ingestion, IoT data, log streaming, and real-time analytics. Event Hub uses a partitioned consumer model with a retention period (1-90 days). Data can be consumed by Azure Stream Analytics, Spark, or custom consumers.

Q169. What is Azure Event Grid?

✔ **Answer:** Event Grid is a fully managed event routing service for reactive/event-driven programming. It distributes events from sources (Blob Storage, Resource Manager, custom apps) to subscribers (Functions, Logic Apps, webhooks). Unlike Event Hub (streaming), Event Grid is for discrete events that need to trigger reactions.

Q170. What is Azure Notification Hubs?

✔ **Answer:** Notification Hubs provides a cross-platform, scalable push notification infrastructure. You send one API call and it handles delivering notifications to millions of devices across iOS (APNs), Android (FCM), Windows (WNS), and other platforms. It abstracts the complexity of managing different platform notification services.

Q171. What is Azure Cognitive Services?

✔ **Answer:** Azure Cognitive Services (now part of Azure AI Services) provides pre-built AI APIs for vision, speech, language, and decision capabilities — without needing ML expertise. Examples: Computer Vision (analyze images), Speech to Text, Translator, Text Analytics (sentiment, key phrases), Face API, and Form Recognizer.

Q172. What is Azure DevOps?

✔ **Answer:** Azure DevOps is a set of developer services for teams to plan, develop, test, and deploy software. Includes: Azure Boards (work tracking), Azure Repos (Git hosting), Azure Pipelines (CI/CD), Azure Test Plans (testing), and Azure Artifacts (package management). Integrates with GitHub and many third-party tools.

Q173. What is Azure Application Gateway WAF?

✔ **Answer:** Application Gateway can be enabled with Web Application Firewall (WAF) to protect web apps from common exploits defined in OWASP ruleset — SQL injection, XSS, CSRF, etc. WAF can be in Detection mode (log only) or Prevention mode (block attacks). You can customize rules to reduce false positives.

Q174. What is Azure Spring Apps?

✔ **Answer:** Azure Spring Apps (formerly Azure Spring Cloud) is a fully managed service for deploying Spring Boot microservices. It handles service discovery, configuration management, scaling, monitoring, and updates — letting developers focus on code. Supported tiers: Basic, Standard, and Enterprise (with VMware Tanzu components).

Q175. What is Azure Web Application Firewall (WAF)?

✔ **Answer:** WAF provides centralized protection for web applications from common exploits. In Azure, WAF is available on Application Gateway (regional WAF for single region), Azure Front Door (global WAF), and Azure CDN. It uses OWASP Core Rule Sets and Microsoft-managed rules. Custom rules allow organization-specific block/allow logic.

Q176. What is Microsoft Defender for Cloud?

✔ **Answer:** Microsoft Defender for Cloud (formerly Azure Security Center + Azure Defender) is a unified security management and threat protection platform. It provides Secure Score (measures your security posture), recommendations to fix vulnerabilities, threat detection across Azure, on-premises, and multi-cloud environments.

Q177. What is Secure Score in Defender for Cloud?

✔ **Answer:** Secure Score is a measurement of your organization's security posture — expressed as a percentage. It reflects how many security recommendations you've implemented. Higher score = more secure. Recommendations are grouped by security controls. Implementing recommendations increases your score and reduces risk.

Q178. What is Azure Key Vault?

✔ **Answer:** Key Vault is a managed service for securely storing and accessing secrets (passwords, connection strings), certificates, and cryptographic keys. Applications access secrets at runtime without storing them in code or config files. Key Vault provides access control via Azure AD, audit logs, and hardware security module (HSM) backing for keys.

Q179. What are the main things stored in Azure Key Vault?

✔ **Answer:** Key Vault stores three types: Secrets (passwords, connection strings, API keys — any text), Keys (RSA or EC cryptographic keys for encryption/signing, optionally HSM-backed), and Certificates (TLS/SSL certs with auto-renewal capabilities). Each item has versioning, expiry settings, and individual access policies.

Q180. What is Microsoft Defender for Endpoint?

✔ **Answer:** Defender for Endpoint is an enterprise endpoint detection and response (EDR) platform for Windows, macOS, Linux, iOS, and Android. It provides threat detection, investigation, automated remediation, vulnerability management, and attack surface reduction. It integrates with Defender for Cloud for Azure VM protection.

Q181. What is Microsoft Sentinel?

✔ **Answer:** Microsoft Sentinel is a cloud-native SIEM and SOAR solution. It collects data from all sources (Azure, on-premises, other clouds), uses AI/ML to detect threats, provides investigation tools (incident timeline, entity mapping), and automates responses via playbooks (Logic Apps). Built on Azure Monitor/Log Analytics.

Q182. What is Azure DDoS Protection?

✔ **Answer:** Azure DDoS Protection Standard provides adaptive tuning using ML to learn your traffic patterns and automatically mitigate volumetric, protocol, and application layer DDoS attacks. It provides real-time attack telemetry, post-attack reports, and cost protection (credits for scale-out during attacks). Deployed per VNet.

Q183. What is Azure Information Protection (AIP)?

✔ **Answer:** AIP (now part of Microsoft Purview Information Protection) classifies and protects documents/emails by applying labels. Labels can be applied manually or automatically based on content. Protection includes encryption, access restrictions, and visual markings. Used to protect sensitive data like financial reports, HR records, and IP.

Q184. What is Azure Defender for Storage?

✔ **Answer:** Microsoft Defender for Storage detects unusual and potentially harmful activity on storage accounts — like access from unusual locations, unusual data exfiltration, or malware uploaded to blob storage. It uses threat intelligence and behavioral analytics to alert on suspicious activities in Azure Storage.

Q185. What is Just-In-Time Access and why is it important?

✔ **Answer:** JIT Access (in Defender for Cloud) locks management ports (RDP/SSH) on VMs by default. Access is granted only when requested, for a specific duration, to specific IP addresses — automatically creating temporary NSG rules. This dramatically reduces the attack surface for brute-force attacks targeting management ports.

Q186. What is Microsoft Defender for SQL?

✔ **Answer:** Defender for SQL monitors Azure SQL databases and SQL Server on VMs for threats like SQL injection, anomalous access patterns, and brute-force attacks. It also performs vulnerability assessment to identify and remediate database misconfigurations. Integrated with Defender for Cloud for unified security management.

Q187. What is the difference between SIEM and SOAR?

✔ **Answer:** SIEM (Security Information and Event Management) collects, correlates, and analyzes security data from multiple sources to detect threats. SOAR (Security Orchestration, Automation, and Response) automates responses to detected threats using playbooks. Microsoft Sentinel combines both — it detects threats (SIEM) and automates responses (SOAR).

Q188. What is Azure Security Benchmark?

✔ **Answer:** Azure Security Benchmark (ASB) is a set of high-impact security recommendations based on common frameworks (CIS, NIST). It maps each recommendation to Azure services and provides guidance on implementation. Defender for Cloud uses ASB as its default policy initiative for assessing your security posture.

Q189. What is Role Separation and least privilege in Azure?

✔ **Answer:** Least privilege means granting users/apps only the minimum permissions needed for their tasks — nothing more. Role separation means different teams have different roles (e.g., security team has Security Admin, developers have Contributor on dev resource groups only). Together, they minimize the blast radius of compromised accounts.

Q190. What is Azure Firewall Premium vs Standard?

✔ **Answer:** Azure Firewall Standard provides L3-L7 filtering, FQDN filtering, threat intelligence. Premium adds TLS inspection (decrypt HTTPS traffic for inspection), IDPS (Intrusion Detection and Prevention System) with 58,000+ signatures, URL filtering, and web categories. Premium is for organizations needing deep packet inspection.

Q191. What is Network Security Group (NSG) vs Azure Firewall?

✔ **Answer:** NSGs are free, basic L4 (TCP/UDP port-based) filters applied at subnet or NIC level — good for simple allow/deny rules between subnets. Azure Firewall is a premium, stateful L7 firewall as a service with FQDN filtering, threat intelligence, TLS inspection, and central management. Use NSGs for basic filtering and Azure Firewall for advanced perimeter security.

Q192. What is HTTPS vs TLS and why does it matter in Azure?

✔ **Answer:** TLS (Transport Layer Security) is the encryption protocol that secures HTTPS. In Azure, you should enforce HTTPS for App Services, Storage, API Management, and other services. Azure Application Gateway, Front Door, and CDN handle SSL/TLS termination. Certificates can be managed via Azure Key Vault with auto-rotation.

Q193. What is Vulnerability Assessment in Azure?

✔ **Answer:** Vulnerability Assessment scans Azure resources for security vulnerabilities and misconfigurations. Defender for SQL uses VA for databases. Defender for Servers uses Qualys or Microsoft's built-in scanner for VMs. Results integrate into Defender for Cloud with recommendations, severity ratings, and remediation guidance.

Q194. What is Azure Private Link from a security perspective?

✔ **Answer:** Private Link ensures traffic between your VNet and Azure PaaS services (Storage, SQL, Key Vault) stays on the Microsoft backbone — never traversing the public internet. This eliminates data exfiltration risk (data can't leave to internet) and allows services to be accessed from on-premises through private connectivity.

Q195. What is Microsoft Entra ID?

✔ **Answer:** Microsoft Entra ID is the new name for Azure Active Directory (Azure AD). Microsoft rebranded all Azure AD products under the 'Microsoft Entra' family in 2023. Entra ID remains the same cloud identity platform — it's the central identity service for Microsoft 365, Azure, and third-party SaaS applications.

Q196. What is Azure Defender for Containers?

✔ **Answer:** Microsoft Defender for Containers (formerly Defender for Kubernetes) provides security for container environments. It detects misconfigurations in Kubernetes clusters, threats at runtime, and vulnerabilities in container images in ACR. It monitors AKS clusters and arc-enabled Kubernetes for suspicious activities.

Q197. What is Certificate Management in Key Vault?

✔ **Answer:** Key Vault can store, manage, and auto-renew TLS/SSL certificates. It integrates with trusted CAs (DigiCert, GlobalSign). When a certificate nears expiry, Key Vault can automatically renew and update it. Applications and services (App Service, Application Gateway) can reference Key Vault certificates without manually managing renewals.

Q198. What is Azure Policy for Security compliance?

✔ **Answer:** Azure Policy enforces security standards across your resources. For example, policies can require all storage accounts to use HTTPS only, require SQL to use Azure AD authentication, deny VMs without disk encryption, or audit Key Vault soft delete settings. Security-related policies reduce the chance of misconfigurations.

Q199. What is Microsoft Defender for Cloud's regulatory compliance dashboard?

✔ **Answer:** The regulatory compliance dashboard in Defender for Cloud shows your compliance status against standards like PCI DSS, HIPAA, ISO 27001, SOC 2, and CIS Benchmarks. Each control in the standard maps to Azure security recommendations. You can see pass/fail status, evidence, and export compliance reports for auditors.

Q200. What is the difference between encryption at rest and encryption in transit?

✔ **Answer:** Encryption at rest protects stored data — Azure Storage, disks, and databases encrypt data at rest using AES-256 by default. Encryption in transit protects data moving over networks — enforced via TLS/HTTPS. Both are required for comprehensive data protection. Key Vault stores the encryption keys for at-rest encryption.

9 Backup & Disaster Recovery

Q201. What is a Recovery Services Vault?

✔ **Answer:** A Recovery Services Vault is a storage entity in Azure used to store backup data and recovery points. It holds backups for Azure VMs, on-premises machines (via MARS agent), SQL/SAP databases, Azure Files, and more. Vault is tied to a region and subscription. It provides backup policies, monitoring, and restore capabilities.

Q202. What is the difference between Azure Backup and Azure Site Recovery?

✔ **Answer:** Azure Backup is for data protection — creating recoverable copies of data to restore from data loss, accidental deletion, or ransomware. Azure Site Recovery (ASR) is for business continuity — replicating entire workloads to a secondary region so you can fail over if the primary region becomes unavailable. Both use Recovery Services Vault.

Q203. What are RPO and RTO?

✔ **Answer:** RPO (Recovery Point Objective) is the maximum acceptable data loss measured in time — how old can your last backup be? RTO (Recovery Time Objective) is the maximum acceptable downtime — how quickly must you restore after a disaster? Lower RPO/RTO = more expensive. Design backup/DR strategy around business-defined RPO/RTO requirements.

Q204. What are the backup options for Azure VMs?

✔ **Answer:** Azure VMs can be backed up using Azure Backup (VM-level backup to Recovery Services Vault — daily/hourly, application-consistent), Azure Site Recovery (for DR, not just backup), VM Snapshots (manual point-in-time disk snapshots), or third-party tools like Veeam. Azure Backup is the recommended native solution.

Q205. What is application-consistent backup?

✔ **Answer:** Application-consistent backup ensures all data in memory and in-flight transactions are flushed to disk before the snapshot is taken. For Windows, this uses VSS (Volume Shadow Copy Service). For Linux, pre/post scripts freeze I/O. This ensures a consistent state that can be cleanly restored without data corruption.

Q206. What is Cross-Region Restore (CRR)?

✔ **Answer:** CRR allows you to restore Azure VM backups in a secondary region (paired region) even if the primary region is healthy. This is useful for DR drills, compliance requirements, or when the primary region is experiencing issues. CRR must be enabled on the Recovery Services Vault and uses GRS-stored backup data.

Q207. What is Soft Delete in Azure Backup?

✔ **Answer:** Soft Delete for Backup protects against accidental or malicious deletion of backup data. When Soft Delete is enabled, deleted backup data is retained for 14 additional days (at no charge) before permanent deletion. This gives you time to recover from ransomware attacks or accidental disable/delete operations.

Q208. What is Azure Backup Center?

✔ **Answer:** Backup Center is a single unified management experience for all backup operations across Azure. From one place, you can monitor backup jobs, manage policies, govern backup estate, analyze costs, and detect anomalies — across all workloads (VMs, SQL, SAP, Files, Blobs) and multiple subscriptions.

Q209. What is Geo-Redundant Storage (GRS) for Recovery Services Vault?

✔ **Answer:** Recovery Services Vaults use GRS by default, replicating backup data to a paired secondary region. This ensures backup data survives regional disasters. LRS (Locally Redundant Storage) keeps data in one region at lower cost — suitable when you have another DR strategy. ZRS (Zone-Redundant Storage) spreads data across availability zones.

Q210. What is Azure Backup MARS Agent?

✔ **Answer:** MARS (Microsoft Azure Recovery Services) Agent backs up files, folders, and system state from on-premises Windows machines and Azure VMs directly to a Recovery Services Vault. It doesn't require a System Center DPM server. MARS supports up to 3 scheduled backups per day and up to 9,999 recovery points.

Q211. What is Instant Restore in Azure VM Backup?

✔ **Answer:** Instant Restore allows you to restore Azure VM files or the entire disk instantly from locally stored snapshots (retained for 1-5 days) without waiting for data to be transferred from the vault. Full vault-based restore takes longer as data must be fetched. Instant Restore is faster for recent recovery points.

Q212. What is Azure Site Recovery replication for Azure VMs?

✔ **Answer:** ASR replicates Azure VMs continuously to a secondary region. Changes are captured as snapshots every few minutes (crash-consistent) and committed (application-consistent) periodically. If you fail over, the replicated VM starts in the secondary region. You can test failover without affecting production by using an isolated test VNet.

Q213. What is Failover and Failback in ASR?

✔ **Answer:** Failover is switching workloads from the primary region to the secondary (DR) region during a disaster. After the primary region recovers, Failback brings workloads back. In ASR, you can do planned failover (zero data loss, for maintenance), unplanned failover (some data loss possible), and test failover (non-disruptive testing).

Q214. What is Azure VM backup policy?

✔ **Answer:** A backup policy defines the schedule and retention for VM backups. You can configure: backup frequency (daily or hourly), time of backup, and retention for daily, weekly, monthly, and yearly points. Policies are assigned to VMs through the Recovery Services Vault. Multiple VMs can share the same policy.

Q215. What is Azure Backup for SQL Server on Azure VMs?

✔ **Answer:** Azure Backup supports workload-aware backup for SQL Server running on Azure VMs. It provides full, differential, and log backups with 15-minute RPO. Backups integrate with SQL Server VSS, enabling application-consistent restores to a specific point in time. Configuration is done through Defender for Cloud or Recovery Services Vault.

Q216. What is tiered backup storage in Azure Backup?

✔ **Answer:** Azure Backup supports tiered storage: Standard Tier (active recovery points in vault storage) and Archive Tier (long-term retention points moved to cool storage at 75% cost savings). You can configure vault policies to move older recovery points to archive tier automatically — ideal for compliance-driven long-term retention.

Q217. What is enhanced policy for Azure VM backup?

✔ **Answer:** Enhanced Policy enables hourly backup for Azure VMs (vs. once-daily in standard policy), supports backup for trusted launch VMs, and provides selective disk backup. It uses snapshot tier for faster restore and supports multiple backups per day for more granular RPO requirements.

Q218. What is Azure Backup for Azure Files?

✔ **Answer:** Azure Backup for Azure Files creates share snapshots of file shares in Azure Storage. Snapshots are instant and storage-efficient (incremental). Restores can be done at the share level or individual file/folder level. Retention can be set daily, weekly, monthly, and yearly. Policy is managed through Recovery Services Vault.

Q219. What is Multi-user Authorization (MUA) in Azure Backup?

✔ **Answer:** MUA adds an extra layer of protection to critical backup operations. You designate a Resource Guard (in a different subscription managed by a security admin). For operations like disabling soft delete, modifying backup policies, or deleting backup data, the backup admin needs permissions on the Resource Guard — preventing single-account compromise.

Q220. What is Azure Business Continuity Center?

✔ **Answer:** Azure Business Continuity Center (newer service) is an evolution of Backup Center that unifies backup and disaster recovery management in one place. It provides governance, monitoring, and management for both Azure Backup and Azure Site Recovery from a single pane — covering data protection and DR for all protected workloads.

10 Governance & Compliance

Q221. What is Azure Governance?

✔ **Answer:** Azure Governance is the set of processes, policies, and tools used to ensure Azure resources are managed consistently and comply with organizational standards and regulatory requirements. Key tools include Azure Policy, RBAC, Management Groups, Blueprints, Cost Management, and Azure Advisor — applied at scale across subscriptions.

Q222. What is the Azure Well-Architected Framework?

✔ **Answer:** The Azure Well-Architected Framework is a set of guiding principles for building high-quality, cloud solutions. It has five pillars: Reliability (resilience, DR), Security (protect assets), Cost Optimization (control spending), Operational Excellence (DevOps, monitoring), and Performance Efficiency (scale, optimize). Use it to review and improve architectures.

Q223. What is the principle of least privilege in Azure?

✔ **Answer:** Least privilege means assigning only the minimum permissions required to perform a task — nothing more. In Azure, this means using specific built-in roles instead of Owner, scoping RBAC assignments at the lowest necessary scope (resource vs. subscription), using Managed Identities over service principals with passwords, and reviewing access regularly.

Q224. What is Compliance Manager in Azure?

✔ **Answer:** Microsoft Purview Compliance Manager helps organizations manage their compliance posture across Microsoft 365, Azure, and multi-cloud. It provides pre-built assessments for 300+ regulations (GDPR, HIPAA, ISO 27001), tracks your compliance score, and provides step-by-step guidance on implementing controls.

Q225. What is Azure Policy 'deny' effect?

✔ **Answer:** When a policy with 'Deny' effect is assigned, it blocks any resource creation or update that doesn't comply with the policy rule. For example, a 'Deny' policy requiring all storage accounts to use HTTPS-only will block creation of storage accounts with HTTP allowed. Deny policies prevent non-compliant resources from being created.

Q226. What is Azure Policy remediation?

✔ **Answer:** Remediation automatically fixes non-compliant existing resources. Policies with 'DeployIfNotExists' or 'Modify' effects can have remediation tasks created to scan and fix already-deployed resources. For example, a policy requiring Diagnostic Settings can remediate by deploying them to existing resources that don't have them.

Q227. What is Cost governance in Azure?

✔ **Answer:** Cost governance involves setting policies and processes to control Azure spending. This includes: tagging all resources for cost allocation, using Azure Budgets to alert on overspending, Azure Policy to require tags and restrict expensive VM SKUs, reservations for predictable workloads, and regular Cost Management analysis to identify waste.

Q228. What is the difference between Azure Compliance and Azure Certification?

✔ **Answer:** Azure Compliance refers to Microsoft's own adherence to regulatory standards — Azure has 100+ compliance certifications (SOC 1/2, ISO 27001, PCI DSS, HIPAA, FedRAMP, etc.). These certifications mean Azure infrastructure meets requirements. Your workloads on Azure still need separate compliance efforts for your own data and configurations.

Q229. What is Azure Trust Center?

✔ **Answer:** Azure Trust Center (now Microsoft Trust Center at trust.microsoft.com) is Microsoft's transparency hub for privacy, security, and compliance. It provides information about Microsoft's compliance certifications, regulatory adherence, data protection practices, and security controls — useful for customers evaluating Azure for regulated industries.

Q230. What is the Shared Responsibility Model in cloud?

✔ **Answer:** The Shared Responsibility Model defines what Microsoft manages and what you manage in Azure. Microsoft is always responsible for physical datacenter security, network hardware, and hypervisor. In IaaS, you manage OS, apps, data, and identity. In PaaS, Microsoft manages OS/runtime too. In SaaS, Microsoft manages almost everything. Data and identity are always your responsibility.

Q231. What is tagging governance in Azure?

✔ **Answer:** Tagging governance ensures resources are consistently tagged for cost allocation, compliance, and management. Azure Policy can enforce required tags (deny untagged resources), inherit tags from resource groups, and apply default tag values. Cost Management uses tags to generate departmental cost reports. Tags are critical for large enterprises.

Q232. What is Resource Graph in Azure?

✔ **Answer:** Azure Resource Graph is a powerful querying service that lets you explore Azure resources across subscriptions at scale using KQL. Unlike the portal's resource list (limited to single subscription), Resource Graph can query millions of resources across your entire tenant in seconds. Useful for compliance auditing and inventory management.

Q233. What is Azure Subscription Design best practice?

✔ **Answer:** Best practices include: use separate subscriptions for Dev/Test/Production, use Management Groups for policy inheritance, use a 'connectivity' subscription for shared networking, implement consistent naming/tagging, apply Azure Policy at management group level, and use Azure Cost Management for per-subscription billing visibility.

Q234. What is GDPR and how does Azure support it?

✔ **Answer:** GDPR (General Data Protection Regulation) is EU law protecting personal data. Azure supports GDPR compliance through: data residency controls (choose region), encryption at rest/in-transit, Data Subject Requests via Purview, audit logs for access, and Microsoft's DPA (Data Processing Agreement). Azure has EU GDPR compliance certifications.

Q235. What is ISO 27001 compliance in Azure?

✔ **Answer:** ISO 27001 is an international standard for information security management systems (ISMS). Azure infrastructure is certified for ISO 27001. For your workloads, Azure Policy provides ISO 27001 initiative (set of policies) to assess your resource configurations against ISO controls — helping demonstrate compliance in audits.

Q236. What is Azure Managed HSM?

✔ **Answer:** Azure Managed HSM (Hardware Security Module) is a fully managed, single-tenant HSM service for protecting cryptographic keys. Unlike Key Vault (software-backed keys by default), Managed HSM uses FIPS 140-2 Level 3 validated HSMs. Used for compliance requirements mandating hardware-protected keys — common in banking and government.

Q237. What is Microsoft Defender for DevOps?

✔ **Answer:** Defender for DevOps provides security posture management for your development pipelines (Azure DevOps, GitHub). It scans IaC templates (Bicep, ARM, Terraform) for misconfigurations before deployment, scans containers in CI/CD, and manages secrets exposure in code. It integrates into Defender for Cloud's overview.

Q238. What is Azure Monitor Compliance Reporting?

✔ **Answer:** Azure Policy's compliance dashboard shows what percentage of your resources comply with assigned policies/initiatives. You can export compliance reports, view per-resource compliance status, view compliance trends over time, and set up alerts for compliance state changes. This is used for auditing and governance reporting.

Q239. What is Defender for Cloud's Free vs Defender plans?

✔ **Answer:** Defender for Cloud Free tier provides basic security recommendations, Secure Score, and asset inventory without cost. Paid Defender plans (per service: Servers, Databases, Storage, Containers, App Service, Key Vault, DNS, Resource Manager) add advanced threat detection, vulnerability assessment, and endpoint integration for each service.

Q240. What is Azure Advisor Security recommendations?

✔ **Answer:** Azure Advisor aggregates security recommendations from Defender for Cloud into its Security category. Recommendations include: enable MFA, remove unused access, enable disk encryption, use RBAC for key vault access, enable soft delete on vaults, and more. Implementing these improves your Secure Score.

Q241. What are the Azure Load Balancing options?

✔ **Answer:** Azure provides four main load balancing services: Azure Load Balancer (L4, TCP/UDP, regional), Application Gateway (L7, HTTP/HTTPS, regional with WAF), Azure Front Door (global L7 with CDN and WAF), and Traffic Manager (DNS-based global routing). Choose based on protocol, scope (regional/global), and features needed.

Q242. When to use Traffic Manager vs Front Door?

✔ **Answer:** Traffic Manager routes DNS queries to healthy endpoints globally — it works with any protocol (HTTP, SMTP, TCP) since it operates at DNS level, not IP level. Front Door is for HTTP/HTTPS only but provides active load balancing with failover, WAF, caching, and SSL offload. Use Traffic Manager for non-HTTP or multi-region DNS routing; Front Door for HTTP with CDN/WAF.

Q243. What is health probe in Azure Load Balancer?

✔ **Answer:** Health probes monitor the health of backend VMs/instances. Load Balancer regularly sends probes (HTTP, HTTPS, TCP) to each backend. If a backend fails to respond, Load Balancer stops sending traffic to it. This ensures traffic only goes to healthy instances. Standard Load Balancer supports HTTPS probes; Basic supports HTTP/TCP.

Q244. What is connection draining (graceful termination)?

✔ **Answer:** Connection draining allows existing connections to complete before a backend instance is removed from the load balancer pool (during updates or scale-in). You configure a drain period (timeout). During this time, new connections don't route to the draining instance, but existing connections continue until the timeout or they complete naturally.

Q245. What is the SLA for Azure VMs in Availability Zones?

✔ **Answer:** Azure guarantees 99.99% uptime SLA for VMs deployed across two or more Availability Zones in the same region. Single VM with Premium SSD gets 99.9% SLA. VMs in Availability Sets get 99.95% SLA. No SLA is provided for single VMs with Standard HDD. Higher redundancy = better SLA = higher cost.

Q246. What is Azure Front Door's origin group?

✔ **Answer:** An Origin Group in Azure Front Door is a group of backend servers (origins) that Front Door load balances traffic across. You configure health probe settings, load balancing settings (latency sensitivity, sample size), and session affinity. Front Door supports weighted routing and priority (active/standby) within an origin group.

Q247. What is session affinity (sticky sessions)?

✔ **Answer:** Session affinity (cookie-based) ensures a user's requests always go to the same backend server during a session. Application Gateway and Front Door support cookie-based session affinity. Load Balancer uses 5-tuple hash by default (client IP, port, server IP, port, protocol) for distribution. Sticky sessions help stateful apps but reduce even distribution.

Q248. What is Azure Traffic Manager failover routing?

✔ **Answer:** Priority routing in Traffic Manager sends all traffic to the primary endpoint. If the primary health check fails, Traffic Manager automatically sends traffic to the next priority endpoint. This enables active/passive failover patterns for multi-region DR. Health checks can be HTTP/HTTPS on a custom path to verify app health.

Q249. What is Standard vs Basic Azure Load Balancer?

✔ **Answer:** Standard Load Balancer supports Availability Zones (zone-redundant), unlimited backend pool size, HTTPS health probes, multiple load balancing rules, outbound rules, and HA ports. Basic is free but limited (no zones, 300 backend VMs max, no HTTPS probes, no outbound rules). Standard is required for production and zone-redundant architectures.

Q250. What is inbound NAT rules in Azure Load Balancer?

✔ **Answer:** Inbound NAT Rules forward specific ports from the load balancer's public IP to specific VMs. For example, forward port 50001 to VM1's port 3389, and port 50002 to VM2's port 3389 — enabling direct RDP to individual VMs behind a shared IP. This is useful for direct management access to specific backends.

Q251. What is HA Ports in Azure Load Balancer?

✔ **Answer:** HA (High Availability) Ports is a Standard Load Balancer feature that load balances ALL TCP/UDP ports simultaneously in a single rule — used for internal network virtual appliances (firewalls, routers). Instead of creating rules for each port, one HA Ports rule covers everything. Only available on internal Standard Load Balancer.

Q252. What is Azure Application Gateway autoscaling?

✔ **Answer:** Application Gateway v2 supports autoscaling — it automatically adjusts the number of gateway instances based on traffic load. You set a minimum and maximum instance count. Autoscaling ensures performance during traffic spikes and reduces cost during quiet periods. V2 also supports zone-redundancy for higher availability.

Q253. What is Global vs Regional load balancing?

✔ **Answer:** Regional load balancers (Azure Load Balancer, Application Gateway) distribute traffic within a single Azure region to VMs/containers. Global load balancers (Traffic Manager, Front Door) route traffic across multiple regions to direct users to the nearest or healthiest regional endpoint. Multi-region deployments use both layers.

Q254. What is a Backend Pool in Azure Load Balancer?

✔ **Answer:** The Backend Pool is the collection of resources (VMs, VMSS, or IP addresses) that receive load-balanced traffic. Traffic is distributed across backend pool members based on the load balancing rule and health probe status. Standard Load Balancer backend pools can contain VMs across availability zones and up to 1000 instances.

Q255. What is Azure Front Door routing rules?

✔ **Answer:** Front Door routing rules match incoming requests to origins. You can route based on HTTP method, URL path patterns (e.g., /api/* → backend API, /static/* → CDN), host headers, or protocol. Rules can also configure caching behavior, URL rewriting, URL redirects, and custom headers. This enables sophisticated L7 routing.

Q256. What is zone-redundant Load Balancer?

✔ **Answer:** A zone-redundant Standard Load Balancer is automatically resilient to zone failures — the load balancer frontend survives the loss of any single zone. The frontend IP is replicated across all three availability zones. Backend VMs can be in any zone. This requires a Standard SKU load balancer with a Standard SKU public IP.

Q257. What is Azure Traffic Manager Geographic routing?

✔ **Answer:** Geographic routing in Traffic Manager directs users to specific endpoints based on their geographic location (country, region, or continent). This is used for data sovereignty (EU users must go to EU endpoint), localization (users get language-specific backends), or regulatory compliance. Traffic Manager uses DNS TTL for routing decisions.

Q258. What is weighted routing in Azure Traffic Manager?

✔ **Answer:** Weighted routing distributes traffic across endpoints based on assigned weights (1-1000). For example, 80% to primary endpoint and 20% to secondary for gradual migration/blue-green deployments. When an endpoint fails health check, Traffic Manager redistributes its share proportionally among remaining healthy endpoints.

Q259. What is multivalue routing in Azure Traffic Manager?

✔ **Answer:** Multivalue routing returns all healthy endpoints in a single DNS response — the client chooses which to use. This is designed for DNS records that can return multiple IPv4/IPv6 addresses. Clients with multivalue-aware implementations can retry using a different endpoint if the first fails, improving availability.

Q260. What is Azure Application Gateway redirect?

✔ **Answer:** Application Gateway supports HTTP to HTTPS redirect, URL path-based redirect, and external URL redirect. You can redirect entire sites or specific paths. Redirect rules can send a 301 (permanent) or 302 (temporary) status code. This is commonly used to force HTTPS and redirect old URLs to new paths.

12 Automation & DevOps

Q261. What is Azure Automation?

✔ **Answer:** Azure Automation provides cloud-based automation and configuration management for Azure and non-Azure environments. Key features: Runbooks (PowerShell/Python automation scripts), Update Management (patch VMs), Change Tracking and Inventory (track software changes), Desired State Configuration (DSC for VM config), and Process Automation.

Q262. What is Update Management in Azure?

✔ **Answer:** Update Management (part of Azure Automation) manages OS updates for Windows and Linux VMs — in Azure, on-premises, or other clouds. It assesses compliance, schedules update deployments, and reports results. It uses Log Analytics data and automation runbooks. Can be managed from Azure Arc for non-Azure machines.

Q263. What is Azure DevOps Pipelines?

✔ **Answer:** Azure Pipelines provides CI/CD pipeline automation for building, testing, and deploying applications to any platform — Azure, other clouds, or on-premises. Pipelines are defined in YAML or classic UI. Stages include build, test, and multiple deployment stages (Dev → Test → Prod) with approvals and gates for controlled releases.

Q264. What is Infrastructure as Code (IaC)?

✔ **Answer:** IaC means managing and provisioning infrastructure through code files rather than manual processes. In Azure, IaC options include ARM Templates (JSON), Bicep (DSL for ARM), Terraform (HashiCorp, multi-cloud), Pulumi (general-purpose languages), and Ansible (configuration management). IaC enables repeatable, version-controlled deployments.

Q265. What is Terraform and how does it work with Azure?

✔ **Answer:** Terraform is an open-source IaC tool by HashiCorp that provisions Azure resources using declarative HCL configuration files. It uses the Azure Provider (azurerm) to interact with Azure Resource Manager API. Terraform maintains a state file tracking deployed resources. Key commands: init, plan (preview changes), apply (deploy), destroy.

Q266. What is Azure Arc?

✔ **Answer:** Azure Arc extends Azure management capabilities to on-premises, other cloud, and edge environments. Arc-enabled servers can use Azure Policy, Azure Monitor, Defender for Cloud, Update Management, and RBAC — just like native Azure VMs. Arc also supports Kubernetes clusters, SQL Server, and PostgreSQL outside Azure.

Q267. What is Desired State Configuration (DSC)?

✔ **Answer:** DSC is a PowerShell-based management platform for ensuring VMs maintain a desired configuration state. You write DSC configurations specifying what software, registry settings, and services should exist. Azure Automation State Configuration hosts DSC configurations and monitors VMs for drift from the desired state.

Q268. What is Azure Runbooks?

✔ **Answer:** Runbooks are automation scripts in Azure Automation. They can be PowerShell, Python, or PowerShell Workflow scripts that automate repetitive tasks like starting/stopping VMs, creating resources, or sending alerts. Runbooks can be scheduled, triggered by webhooks, or called from alerts. They run on Azure or on-premises Hybrid Runbook Workers.

Q269. What is Azure DevTest Labs?

✔ **Answer:** DevTest Labs allows you to create lab environments for development and testing quickly and cost-efficiently. It provides self-service provisioning of VMs with pre-configured images, formulas, and artifacts. Labs include cost controls (auto-shutdown, spending caps) and claimable pools. It reduces dev/test costs and streamlines lab management.

Q270. What is GitHub Actions for Azure?

✔ **Answer:** GitHub Actions provides CI/CD automation directly in GitHub repositories. Microsoft provides official Azure Actions for deploying to App Service, AKS, Container Instances, Functions, and more. You authenticate using Azure service principal or federated identity (OIDC — no secret needed). Actions integrate with Azure environments for approvals.

Q271. What is Managed Identities for automation?

✔ **Answer:** Managed Identities are the recommended way for automation scripts/runbooks to authenticate to Azure services. Instead of storing credentials in runbooks, the Automation Account uses its system-assigned managed identity. You grant the identity RBAC permissions on target resources. Azure handles credential rotation automatically — no secrets to manage.

Q272. What is Azure Migrate?

✔ **Answer:** Azure Migrate is a hub of tools for discovering, assessing, and migrating on-premises workloads to Azure. It supports server migration (Hyper-V, VMware, physical), database migration (SQL Server → Azure SQL), web app migration (IIS → App Service), and data migration. It provides dependency mapping and TCO/cost estimates.

Q273. What is Azure Database Migration Service?

✔ **Answer:** Azure Database Migration Service (DMS) facilitates migrations of databases to Azure with minimal downtime. Supports: SQL Server to Azure SQL Database/Managed Instance, MySQL to Azure Database for MySQL, PostgreSQL to Azure Database for PostgreSQL, and MongoDB to Cosmos DB. Online migrations use CDC for near-zero downtime.

Q274. What is Continuous Integration (CI) in Azure DevOps?

✔ **Answer:** CI automatically builds and tests code every time a developer commits changes. In Azure Pipelines, CI pipelines compile code, run unit tests, perform code analysis, and build container images. Fast feedback on code quality prevents bugs from accumulating. CI triggers on pull requests or branch commits.

Q275. What is Continuous Delivery (CD) vs Continuous Deployment?

✔ **Answer:** Continuous Delivery ensures code is always in a deployable state and can be released to production at any time — but release requires manual approval. Continuous Deployment goes further — every passed CI build automatically deploys to production without human intervention. CD in Azure Pipelines supports both via environment approvals.

Q276. What is Azure Pipelines Environments?

✔ **Answer:** Environments in Azure Pipelines represent deployment targets (Dev, Test, Staging, Prod). They track deployment history, allow manual approval gates before deployment, and support checks (Azure Policy, work item linking). You can use resource types (Kubernetes, Virtual Machines) to track individual target machines in an environment.

Q277. What is Bicep vs Terraform for Azure IaC?

✔ **Answer:** Bicep is Azure-native (only Azure), has first-class Azure support with latest API versions immediately, simpler syntax than ARM JSON, and is maintained by Microsoft. Terraform is multi-cloud (Azure, AWS, GCP), has a rich provider ecosystem, mature tooling, and a large community. For Azure-only teams: Bicep. For multi-cloud: Terraform.

Q278. What is Azure Resource Manager Deployment Modes?

✔ **Answer:** ARM supports two deployment modes: Incremental (default — only adds/updates resources defined in template, doesn't delete existing resources not in template) and Complete (deletes resources in the resource group that aren't in the template). Complete mode is powerful but risky — it can delete resources if template is incomplete.

Q279. What is Azure Automation Hybrid Runbook Worker?

✔ **Answer:** Hybrid Runbook Worker extends Azure Automation to run runbooks on machines inside your own datacenter or other clouds. This allows automation to reach on-premises resources (databases, file servers) that aren't accessible from Azure. Workers are installed on Windows or Linux machines and connect to Azure Automation via HTTPS outbound.

Q280. What is the Azure Resource Manager (ARM) API?

✔ **Answer:** The ARM REST API is the foundation that all Azure management tools use. The portal, CLI, PowerShell, and SDKs all call the ARM API. Resource operations follow standard HTTP verbs: GET (read), PUT (create/update), DELETE (remove), PATCH (modify). Understanding the API structure helps when building custom automation or troubleshooting tool issues.

Q281. What is Azure SQL Database?

✔ **Answer:** Azure SQL Database is a fully managed PaaS relational database based on the latest stable SQL Server engine. Microsoft handles infrastructure, patching, backups, and high availability. It supports two purchasing models: DTU (pre-defined bundles of compute+storage) and vCore (separate compute, storage, and memory selection). Tiers: General Purpose, Business Critical, Hyperscale.

Q282. What is the difference between Azure SQL Database and SQL Managed Instance?

✔ **Answer:** SQL Database is a single database or elastic pool — PaaS with most features but no SQL Agent, limited cross-database queries, no CLR. SQL Managed Instance is a nearly 100% compatible SQL Server hosted in Azure inside a VNet — supports SQL Agent, CLR, cross-database queries, and linked servers. MI is better for lift-and-shift migrations.

Q283. What is Azure SQL Elastic Pool?

✔ **Answer:** Elastic Pools let multiple databases share a single pool of resources (eDTUs or vCores). Databases in the pool can use resources as needed up to the pool limit. This is cost-effective when databases have variable, unpredictable usage — they don't all peak at the same time, so shared resources reduce total cost vs. individual allocations.

Q284. What is Azure SQL Hyperscale?

✔ **Answer:** Hyperscale is a vCore service tier for very large OLTP databases (up to 100 TB). It uses a log-based architecture with separate compute and storage, fast backups (snapshot-based, regardless of size), fast scale-up/down, and readable secondary replicas. Hyperscale decouples compute from storage scaling, enabling independent scaling of each.

Q285. What is Azure Database for MySQL/PostgreSQL?

✔ **Answer:** Azure Database for MySQL and PostgreSQL are fully managed open-source relational database services. Azure handles patching, backups, HA, and security. They support Flexible Server deployment option — giving more configuration control, zone-redundant HA, and maintenance window control. Single Server is older and being retired.

Q286. What is Azure Cosmos DB?

✔ **Answer:** Azure Cosmos DB is Microsoft's globally distributed, multi-model NoSQL database. It supports multiple APIs: SQL (document), MongoDB, Cassandra, Gremlin (graph), and Table. Key features: 99.999% availability SLA, single-digit millisecond latency, automatic indexing, and tunable consistency levels (Strong, Bounded Staleness, Session, Consistent Prefix, Eventual).

Q287. What are Cosmos DB consistency levels?

✔ **Answer:** Strong: reads always return latest committed write (highest consistency, highest latency/cost). Bounded Staleness: reads lag behind writes by configured bound (time or versions). Session: consistent within a session — most popular for single-user scenarios. Consistent Prefix: reads never see out-of-order writes. Eventual: lowest latency, highest throughput, weakest consistency.

Q288. What is Azure SQL Database Geo-Replication?

✔ **Answer:** Active Geo-Replication creates up to 4 readable secondary databases in different regions. If the primary fails, you manually failover to a secondary. Secondaries can be used for read workloads to offload the primary. Unlike failover groups, geo-replication requires manual failover and doesn't have an automatic failover policy.

Q289. What is Azure SQL Failover Group?

✔ **Answer:** Failover Groups build on geo-replication and provide a unified connection string (read-write listener and read-only listener) with automatic failover capability. Applications connect using the listener name — if the primary fails, automatic failover promotes the secondary, and the listener automatically points to the new primary. No connection string change needed in app.

Q290. What is Azure SQL Database Serverless?

✔ **Answer:** Azure SQL Database Serverless is a compute tier that automatically scales compute based on workload demand and pauses (bills only for storage) when inactive. It's ideal for intermittent, unpredictable workloads like development/test databases or apps with variable usage. Auto-pause delay configures how long before the DB pauses after inactivity.

Q291. What is Azure Cache for Redis and when to use it?

✔ **Answer:** Azure Cache for Redis is an in-memory data store based on open-source Redis. Use it for: database query result caching (reduce load on SQL), session state storage (stateless app tiers), real-time leaderboards, pub/sub messaging, and rate limiting. Redis stores data in memory for microsecond response times — much faster than database queries.

Q292. What is Azure SQL Transparent Data Encryption (TDE)?

✔ **Answer:** TDE encrypts Azure SQL Database, SQL Managed Instance, and Azure Synapse databases at rest automatically. Data files, log files, and backups are encrypted using AES-256. TDE is enabled by default with service-managed keys. You can use customer-managed keys (CMK) in Azure Key Vault for BYOK (Bring Your Own Key) compliance scenarios.

Q293. What is Azure SQL Dynamic Data Masking?

✔ **Answer:** Dynamic Data Masking limits sensitive data exposure to non-privileged users. You define masking rules on columns (e.g., email shows 'aXXX@XXXX.com', credit card shows 'XXXX-XXXX-XXXX-1234'). The data isn't changed in the database — masking happens at query time for specified users. Privileged users see unmasked data.

Q294. What is Always Encrypted in Azure SQL?

✔ **Answer:** Always Encrypted protects sensitive data (PII, financial data) so that it's encrypted at the client side — the database server NEVER sees plaintext data. Encryption keys are stored in Azure Key Vault or Windows Certificate Store — separate from the database. Only applications with key access can read/write plaintext. Protects against DBA access.

Q295. What is Azure SQL Row-Level Security?

✔ **Answer:** Row-Level Security (RLS) enables access control at the row level in a table — different users see different rows from the same table. You create security policies with filter predicates and block predicates. Example: sales reps can only see their own customers' records, while managers see all. RLS logic lives in the database, not application code.

Q296. What is Azure SQL Database Auditing?

✔ **Answer:** SQL Database Auditing tracks database events and writes them to an audit log — in Azure Storage, Log Analytics, or Event Hub. Auditing helps meet regulatory compliance and understand database activity (logins, queries, schema changes, data access). Extended Events can be used for detailed, low-overhead query-level auditing.

Q297. What is the difference between DTU and vCore purchasing in Azure SQL?

✔ **Answer:** DTU (Database Transaction Unit) bundles compute, memory, and I/O into pre-set packages (10, 50, 100 DTUs, etc.) — simpler but less transparent. vCore lets you independently choose vCores, memory, and storage, choose hardware generation, and understand costs. vCore supports Azure Hybrid Benefit and is recommended for new deployments.

Q298. What is Azure Synapse Analytics?

✔ **Answer:** Azure Synapse Analytics (formerly SQL Data Warehouse) is an integrated analytics platform combining big data and data warehousing. It supports dedicated SQL pools (MPP data warehouse), serverless SQL pools (query data lake without loading), Apache Spark pools, and Data Integration (pipelines). All managed in Azure Synapse Studio.

Q299. What is Azure Data Factory?

✔ **Answer:** Azure Data Factory (ADF) is a cloud-based ETL and data integration service. It enables you to build data pipelines that ingest data from 90+ connectors (databases, files, REST APIs, SaaS apps), transform it using data flows or external compute (Databricks, HDInsight), and load it to destinations. Pipelines can be scheduled or event-triggered.

Q300. What is Azure SQL Managed Instance Business Critical tier?

✔ **Answer:** Business Critical tier uses Always On Availability Groups under the hood to provide a built-in read replica (readable secondary) at no additional cost. It uses local SSD storage for ultra-low latency I/O (1-2ms). It provides the highest IOPS, lowest latency, and built-in HA/DR — suitable for mission-critical OLTP workloads.